# Block practice · Better algorithms, decomposition, abstraction and models

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Decomposition and abstraction

Simulations and models: why we use them and their limits

Make an algorithm faster or work for any amount of data

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# Practice workbook · Decomposition and abstraction

Computational thinking for problem analysis: recognising, defining or explaining decomposition and abstraction in a scenario (planning comments that split a task, a library routine that hides detail).

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-13

# 1. Question 4 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

**4** Eli is designing a three-throw dice game to play on a computer.

He uses abstraction and decomposition to help him.

(a) **Figure 5** shows how Eli uses comments to plan code before writing it.

Eli's preferred language uses a double-slash // to write comments.

```
1  // roll dice 3 times
2
3     // display number rolled
4
5     // update subtotal
6
7     // check for roll of 3 or 6 and increment booster
8
9  // calculate and output final score
10
```

**Figure 5**

State why this is a type of decomposition.

**(1)**

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

**(2)**

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## 1. Question 4 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

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(c) Eli's program has these requirements.

- A roll of the dice is a random number between 1 and 6.
- The dice is rolled three times.
- The numbers rolled are added to a subtotal.
- The final score is calculated by multiplying the subtotal by a booster value.
- The booster value starts at 1.
- Each time a 3 or 6 is rolled, the booster value is incremented by 1.
- The game outputs the value of each roll, and the final score.

**Figure 6** shows a trace table for one execution of the program.

| Output | Subtotal | Booster |
|---|---|---|
| Rolled: 3 | 0 + 3 = 3 | 3 rolled, so incremented to 2 |
| Rolled: 2 | 3 + 2 = 5 | Neither 3 nor 6 rolled |
| Rolled: 6 | 5 + 6 = 11 | 6 rolled, so incremented to 3 |
| Score 33 | Subtotal (11) * Booster (3) = 33 | |

**Figure 6**

Open **Q04c** in the code editor.

Write the code to create this program.

You must use the layout, variables and constants given in **Q04c**.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

**(8)**

**(Total for Question 4 = 11 marks)**

## 1. Question 4 · part b · 2 marks · Working and answer

Working and answer

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## 2. Question 1 · part a · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a only.

**1** Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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P 7 5 7 3 7 A 0 2 1 6

Working and answer

## 3. Question 4 · part a · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part a only.

**4** Eli is designing a three-throw dice game to play on a computer.

He uses abstraction and decomposition to help him.

(a) **Figure 5** shows how Eli uses comments to plan code before writing it.

Eli's preferred language uses a double-slash // to write comments.

```
1  // roll dice 3 times
2
3     // display number rolled
4
5     // update subtotal
6
7     // check for roll of 3 or 6 and increment booster
8
9  // calculate and output final score
10
```

**Figure 5**

State why this is a type of decomposition.

**(1)**

(b) Eli decides to use a library subprogram to generate random numbers that model a dice throw.

Explain why this is an example of abstraction.

**(2)**

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10

P 7 8 6 3 9 A 0 1 0 2 0

## 3. Question 4 · part a · 1 mark · Working and answer

Working and answer

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# Practice workbook · Simulations and models: why we use them and their limits

Why scientists and others use computer simulations, their benefits, problems with relying on them, and their processing demands.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-14

# 1. Question 4 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

**4** Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

**(2)**

(b) Identify the computational model in which several computer systems work together to achieve a goal.

**(1)**

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

**(2)**

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## 1. Question 4 · part a · 2 marks · Working and answer

Working and answer

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## 2. Question 4 · part a · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a only.

**4** Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

**(1)**

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

**(2)**

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

**(3)**

| R | S | W | NOT S | R AND W | (NOT S) OR (R AND W) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

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## 2. Question 4 · part a · 1 mark · Working and answer

Working and answer

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## 3. Question 4 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

**4** Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

**(1)**

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

**(2)**

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

**(3)**

| R | S | W | NOT S | R AND W | (NOT S) OR (R AND W) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

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10

P 6 6 4 8 9 A 0 1 0 2 0

## 3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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## 4. Question 3 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

**3** Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

**(2)**

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

**(1)**

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

**(2)**

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8

P 7 2 9 3 7 A 0 8 2 4

## 4. Question 3 · part a · 2 marks · Working and answer

Working and answer

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# Practice workbook · Make an algorithm faster or work for any amount of data

Judging which of two working algorithms is more efficient, or improving a working algorithm so it copes with any amount of data (array instead of separate variables, LENGTH instead of a hard-coded count).

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-05

# 1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

**4** Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

(ii) State where a **local** variable is accessible.

(1)

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

**Figure 4** and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

**Figure 4**

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P 7 5 7 3 7 A 0 8 1 6

# 1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.

```
1   SEND 'Enter an integer: ' TO DISPLAY
2   RECEIVE theNumber FROM (INTEGER) KEYBOARD
3   IF (theNumber < 40) THEN
4       SEND 'Low range' TO DISPLAY
5   END IF
6   IF (theNumber > 39) AND (theNumber < 70) THEN
7       SEND 'Middle range' TO DISPLAY
8   END IF
9   IF (theNumber > 69) THEN
10      SEND 'High range' TO DISPLAY
11  END IF
```

**Figure 5**

Explain the reason that the algorithm in **Figure 4** is more efficient than the algorithm in **Figure 5**.

(2)

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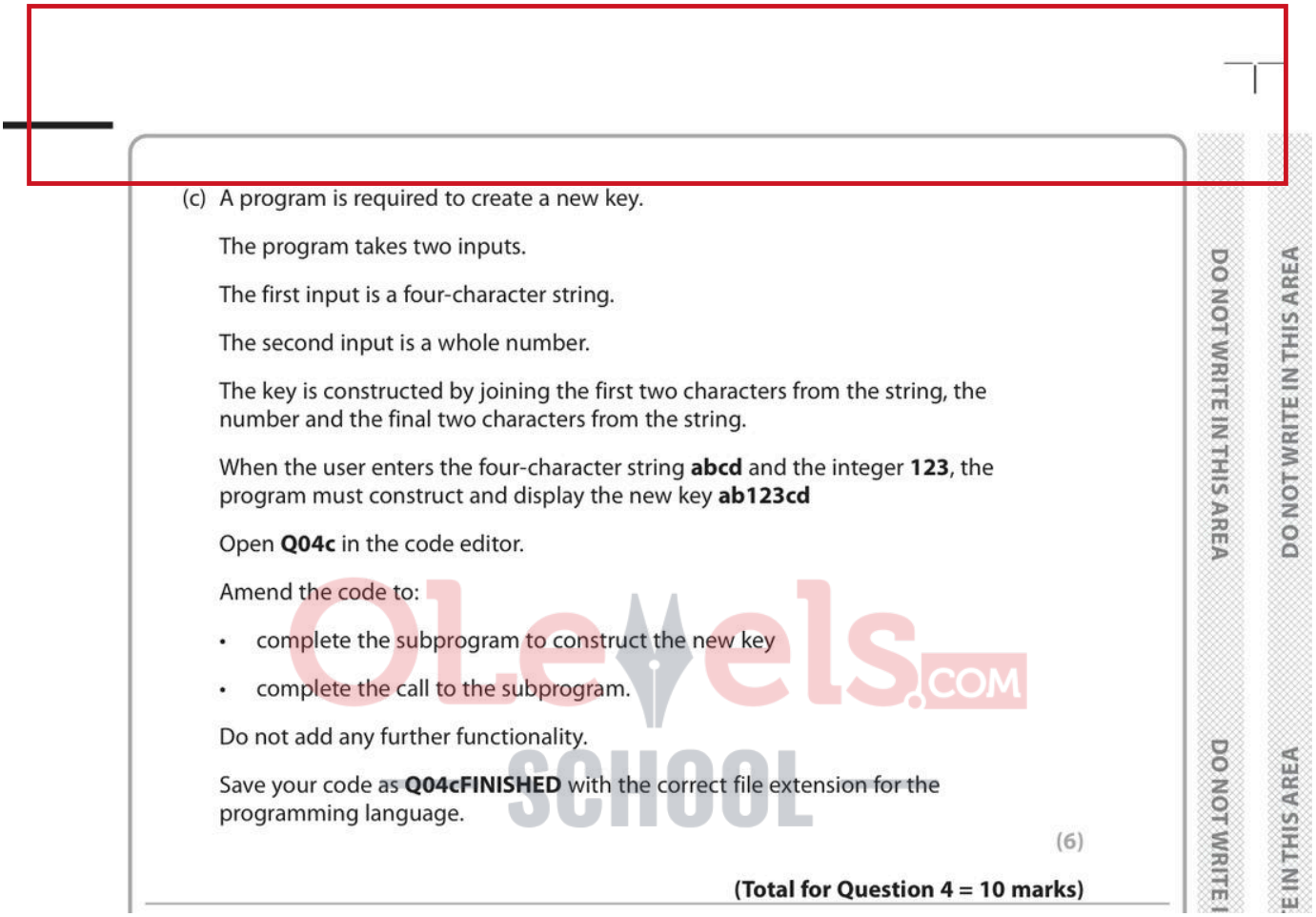
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# 1. Question 4 · part b · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part b only.



(c) A program is required to create a new key.

The program takes two inputs.

The first input is a four-character string.

The second input is a whole number.

The key is constructed by joining the first two characters from the string, the number and the final two characters from the string.

When the user enters the four-character string **abcd** and the integer **123**, the program must construct and display the new key **ab123cd**

Open **Q04c** in the code editor.

Amend the code to:

- complete the subprogram to construct the new key
- complete the call to the subprogram.

Do not add any further functionality.

Save your code as **Q04cFINISHED** with the correct file extension for the programming language.

(6)

**(Total for Question 4 = 10 marks)**

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Working and answer

## 2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

**6** A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

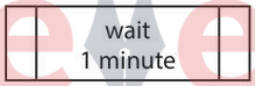
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.




**(6)**

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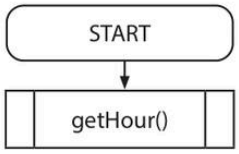
## 2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Draw your flowchart here.




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## 2. Question 6 · part b.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.ii only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

**Figure 4** shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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**Figure 4**

(i) Give the output produced by the algorithm.

**(2)**

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

**(1)**

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## 2. Question 6 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

(iii) State how the pseudocode needs to be changed to make this amendment.

(1)

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**QUESTION 6(c) BEGINS ON THE NEXT PAGE.**

## 2. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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## 3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.iii only.

**6** A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

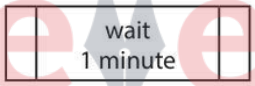
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.




**(6)**

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## 3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

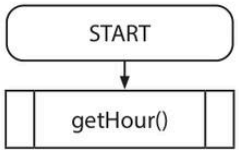
Further practice · Attempt part b.iii only.

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Draw your flowchart here.




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# 3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.iii only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

**Figure 4** shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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**Figure 4**

(i) Give the output produced by the algorithm.

**(2)**

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

**(1)**

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## 3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.iii only.

(iii) State how the pseudocode needs to be changed to make this amendment.

**(1)**

.....

.....

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**QUESTION 6(c) BEGINS ON THE NEXT PAGE.**

## 3. Question 6 · part b.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part b.iii only.

(c) The algorithms will be coded in a high-level language.

Programs written in high-level languages must be translated in order to be executed by a computer.

Discuss the similarities and differences between a compiler and an interpreter.

**(6)**

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**(Total for Question 6 = 16 marks)**

## 3. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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# 4. Question 5 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

**5** Programmers refine algorithms, use searches and read data from files.

(a) **Figure 6** shows an algorithm that determines whether the animal name inputted by the user is found.

```
1  SET a1 TO 'alpaca'
2  SET a2 TO 'bear'
3  SET a3 TO 'camel'
4  SET a4 TO 'deer'
5
6  SEND ('Enter an animal: ') TO DISPLAY
7  RECEIVE target FROM (STRING) KEYBOARD
8
9  IF (a1 = target) THEN
10     SEND ('Found') TO DISPLAY
11 ELSE
12     IF (a2 = target) THEN
13         SEND ('Found') TO DISPLAY
14     ELSE
15         IF (a3 = target) THEN
16             SEND ('Found') TO DISPLAY
17         ELSE
18             IF (a4 = target) THEN
19                 SEND ('Found') TO DISPLAY
20             END IF
21         END IF
22     END IF
23 END IF
```

**Figure 6**

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DO NOT WRITE IN THIS AREA



P 7 5 7 3 7 A 0 1 2 1 6

## 4. Question 5 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

Explain **one** improvement to this algorithm that will reduce the number of variables required and will enable it to work with any number of animals.

(2)

(b) Binary search is a divide and conquer algorithm.

Here is a list of data.

| 10 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|

A binary search is used to check whether the number 11 is in the list.

The midpoint is calculated by adding the high index to the low index and **integer dividing** by 2

Complete the table to show the values for the high index, the low index and the midpoint on each pass of a binary search.

You may not need to use all the rows in the table.

(4)

| high index | low index | midpoint |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

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## 4. Question 5 · part a · 2 marks · Working and answer

Working and answer

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